Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: Phase I, Open-label, Multicenter Study to Evaluate the Imaging Performance, Safety, Biodistribution and Dosimetry of [68Ga]-FF58 in Adult Patients With Selected Solid Tumors Expected to Overexpress αvβ3 and αvβ5 Integrins **Short Title/Acronym:** Study of [68Ga]-FF58 in Patients With Selected Solid Tumors Expected to Overexpress αvβ3 and αvβ5 Integrins **Protocol Number:** CAAA504A12101 **NCT Number:** NCT04712721 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [68Ga]-FF58 (Targeted PET Radiotracer) **Phase of Development:** Early Phase 1 / Phase I (First-In-Human) **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the imaging properties, safety, biodistribution, and dosimetry properties of the radiotracer [68Ga]-FF58 in adults with selected solid tumors expected to overexpress alpha-v beta 3 (αvβ3) and alpha-v beta 5 (αvβ5) integrins. **Secondary Objectives:** To evaluate the number of participants with Treatment-Emergent Adverse Events (TEAEs) and assess the targeting properties via Tumor-to-Background Ratios (TBR) by lesion location. **Optimization Target:** αvβ3 and αvβ5 Integrin Targeting and Identification.

2.2 Background & Rationale To investigate [68Ga]-FF58 as a novel positron emission tomography (PET) imaging agent. The αvβ3 and αvβ5 integrin pathways play a crucial role in tumor angiogenesis and progression in aggressive solid tumors (such as glioblastoma, breast cancer with brain metastases, and gastrointestinal cancers). This study evaluates the capacity of [68Ga]-FF58 to effectively target and image these specific integrins in vivo.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Imaging study) **Control Type:** Single-arm (Parallel assignment across tumor cohorts) **Blinding:** Open-label **Randomization:** Non-Randomized **Medical Methodology:** PET/CT Imaging and Radiotracer Dosimetry

3.2 Planned Enrollment & Duration Target N: 80 patients planned (24 for the imaging characterization part; trial was terminated early due to business reasons) **Number of Sites:** Multicenter

Estimated Study Start: January 2021 **Estimated Primary Completion:** Early Terminated **Total Duration:** Short-term safety follow-up (study early terminated).

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent must be obtained prior to participation. 2. Histologically or cytologically confirmed locally advanced or metastatic progression of: relapsed/refractory (r/r) Glioblastoma Multiforme (GBM) progressed after radiation without prior bevacizumab; OR Breast Cancer (BC) metastasized to the brain; OR Gastroesophageal Adenocarcinoma (GEA); OR Pancreatic Ductal Adenocarcinoma (PDAC).

4.2 Exclusion Criteria 1. Creatinine clearance (calculated using Cockcroft-Gault formula) < 40 mL/min. 2. QTcF > 480 msec on screening ECG or congenital long QT syndrome. 3. Any condition requiring chronic treatment with anticoagulants or antiplatelet agents, or a known bleeding disorder. 4. Pregnancy or administration of another radiopharmaceutical within 10 half-lives of its radionuclide prior to injection.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Management Pathway: Targeted radiotracer administration and imaging pathway. **Dosage/Frequency:** Single dose administration of [68Ga]-FF58 (150 MBq to 250 MBq, or approx 3 MBq/Kg). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Single administration on Day 1, followed by PET/CT imaging at different time points (4 scans).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for underlying advanced malignancies. **Prohibited:** Chronic treatment with anticoagulants or antiplatelet agents.

5.3 Education Protocols Education Protocol (HCP): Radiation safety, radiopharmaceutical handling, and PET/CT acquisition protocol. **Education Protocol (Patient):** Radiation exposure safety instructions (e.g., breastfeeding limits, abstinence protocols post-dose).

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6. Study Timeline & Visit Schedule

Onsite Visit Cadence: Screening, Day 1 (Imaging), Safety Follow-up. **Checklist Review Interval:** N/A (Single administration focus).

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Prior to Day 1	Informed Consent, Medical History, ECG (QTcF), Renal/Coagulation Labs
Baseline / Imaging	Day 1	Single IV dose of [68Ga]-FF58, PET/

CT imaging acquired at 4 different timepoints | | **End of Study** | Follow-up | Safety evaluations, TEAE monitoring, resolution of acute reactions |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time-activity curves (TACs) from [68Ga]-FF58 PET/CT images, evaluated as mean Standard Uptake Values (SUV) across various organs to characterize biodistribution. **Measurement Method:** [68Ga]-FF58 PET imaging acquired on Day 1, paired with conventional imaging (CT/MRI).

7.2 Secondary & Exploratory Endpoints * Number of participants with Treatment-Emergent Adverse Events (TEAEs) from Day 1 to last follow-up. * Tumor to Background Ratio (TBR) calculated as SUVmax(lesion) / SUVmean for the reference region.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of TEAEs starting from the first single dosing on Day 1 up to the last follow-up visit or consent withdrawal. **Serious Adverse Events (SAEs):** Monitored through clinical and laboratory safety parameters (TESAEs and deaths due to AEs). **Remote Monitoring Frequency**: As needed for TEAE resolution. **Data Monitoring Committee (DMC)**: Internal safety monitoring standard for Novartis First-In-Human radiotracer studies.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received the radiotracer) and Imaging/Dosimetry Evaluable Set. **Sample Size Justification**: Planned for $N=80$ total (20 per tumor type) to adequately capture SUV/TBR variations and pharmacokinetic distribution across specific αvβ3/αvβ5 expressing cohorts. **Handling of Missing Data**: Standard imputation methodologies. Due to the early termination of the study (business reasons), the expansion phase was not initiated.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite clinical and imaging center monitoring to ensure proper radioactive material handling, dosing protocol adherence, and PET/CT calibration. **Quality Audit Mechanism & Trail:** eCRF locking, centralized review of PET/CT scan acquisitions, and precise radiodosimetry modeling logs/metrics.

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11. Study Identifiers & Governance

- **Primary ID:** CAAA504A12101
- **Registry Number:** NCT04712721
- **Sponsor/Lead Organization:** Novartis Pharmaceuticals
- **Study Title:** Study of [68Ga]-FF58 in Patients With Selected Solid Tumors Expected to Overexpress αvβ3 and αvβ5 Integrins
- **Official Regulatory Title:** Phase I, Open-label, Multicenter Study to Evaluate the Imaging Performance, Safety, Biodistribution and Dosimetry of [68Ga]-FF58 in Adult Patients With Selected Solid Tumors Expected to Overexpress αvβ3 and αvβ5 Integrins

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12. Clinical Framework (Solid Tumors / Imaging Specific)

- **Primary Condition:** Solid Tumors (GBM, BC with brain mets, GEA, PDAC)
- **Diagnosis Threshold:** Histologically or cytologically confirmed locally advanced or metastatic progression
- **Medical Methodology:** PET/CT Imaging and Radiotracer Dosimetry
- **Optimization Target:** αvβ3 and αvβ5 Integrin Targeting and Identification

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13. Study Procedures & Methodology

- **Management Pathway:** Targeted radiotracer administration and imaging pathway
- **Education Protocol (HCP):** Radiation safety, radiopharmaceutical handling, and PET/CT acquisition protocol
- **Education Protocol (Patient):** Radiation exposure safety instructions (e.g., breastfeeding limits, abstinence protocols post-dose)
- **Quality Audit Mechanism:** Centralized review of PET/CT scans and radiodosimetry metrics

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14. Observation & Follow-Up Schedule

- **Total Duration:** Short-term safety follow-up (study early terminated)
- **Onsite Visit Cadence:** Screening, Day 1 (Imaging), Safety Follow-up
- **Remote Monitoring Frequency:** As needed for TEAE resolution
- **Checklist Review Interval:** N/A (Single administration focus)

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]